



**IN
PARTNERSHIP
WITH
PLYMOUTH
UNIVERSITY**

**PUSL3122 HCI, Computer
graphics and visualisation**

Topic covered this lecture (Lecture 7)

Introduction to Data visualisation

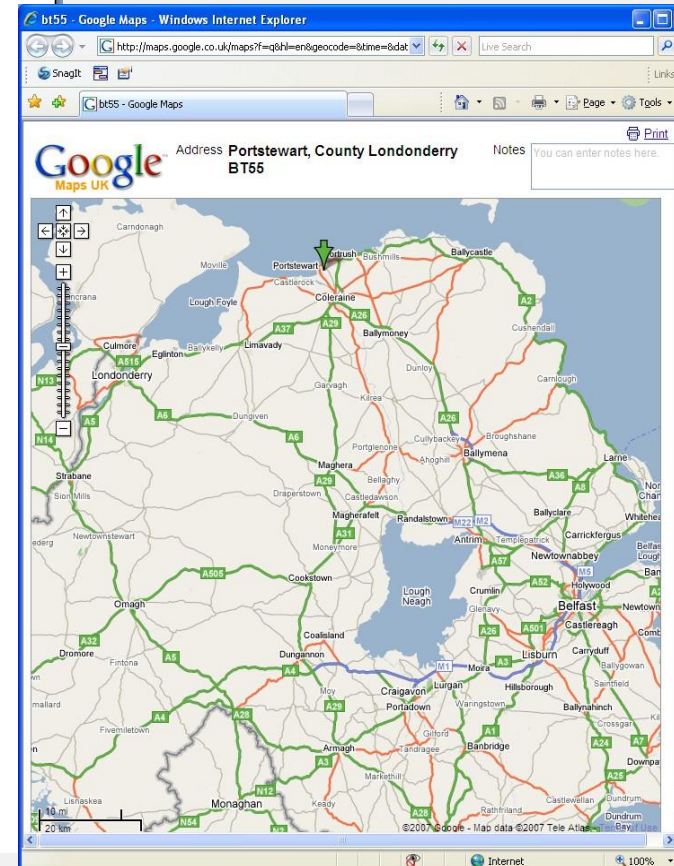
1. Introduction to Visualization

- Visualization – a **mental** process
- Visual adj. (Oxford English Dictionary)
 - Of or used in seeing
- Visualize v. (also –ise) (-zing or –sing)
 - Imagine visually
- Definition by Vandoni 1989
 - “the process of **transforming information** into a visual form, enabling users to **observe** the information. The resulting visual display enables the scientist or engineer to **perceive** visually **features** which are **hidden** in the data but nevertheless are needed for **data analysis** and exploration”

Belfast Airport to Portstewart – Prefer text or a map?

1. Head northeast on A2/Harbour Rd toward The Hill
Continue to follow A2 0.3 mi
2. Slight right at B185/Station Rd
Go through 1 roundabout 2.8 mi
5 mins
3. Turn left at B185/Ring Rd 0.3 mi
1 min
4. At Portrush Road Roundabout, take the 2nd exit onto A29/Ring Rd
Go through 2 roundabouts 2.7 mi
6 mins
5. At Lodge Road Roundabout, take the 1st exit onto A26/New Bridge Rd
Go through 3 roundabouts 22.5 mi
31 mins
6. Slight left at M2 (signs for Belfast/Larne/Antrim/M2) 4.5 mi
4 mins
7. At Larne Road Roundabout, take the 3rd exit onto A26/Ballee Rd E heading to Belfast/Antrim
Continue to follow A26 8.3 mi
10 mins
8. At Dunsilly Roundabout, take the 1st exit onto the M2 ramp to Belfast/Larne 0.4 mi
1 min
9. Merge onto M2 14.7 mi
14 mins
10. Take the exit onto M2 2.7 mi
3 mins
11. Continue on M3/M3 Cross Harbour (signs for Belfast (E)/Newcastle/Bangor/M3)
Continue to follow M3 Cross Harbour 1.0 mi
1 min
12. Continue on A2 2.6 mi
4 mins
13. Take the exit toward Airport Road West Mini Roundabout 0.2 mi
14. At Airport Road West Mini Roundabout, take the 1st exit onto Airport Road West Link 0.1 mi
15. Enter Airport Road West Roundabout 272 ft

These directions are for planning



<http://maps.google.co.uk/>

Visualization umbrella



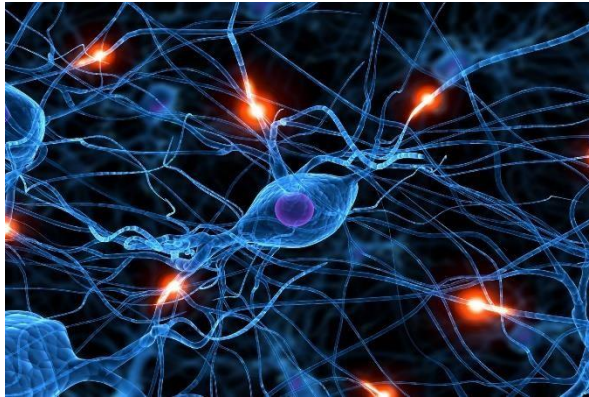
From the **literature** the **scope** of visualisation is clearly diverse and incorporates, or intersects with, nine substantial areas of computer science research and development:

- animation
- software visualisation
- visualisation environments
- volume rendering and visualization
- computer graphics
- visual programming
- image processing
- virtual reality
- **Scientific Visualisation** (interesting)
- **Information Visualization** (approx. 35 years)

*These may apply to
data, program
execution or code*

Is visualization needed?

Teaching



<https://humanities.drury.edu/wp-content/uploads/2014/11/neural-synapse.jpg>

Entertainment



<https://orlandoinformer.com/universal/shrek-4-d/>

Design & Modelling



Slide 51

<http://www.sharc.co.uk/>

Laser guided surgery



<http://groups.csail.mit.edu/vision/medical-vision/surgery/surgical-navigation.html>

What can you see?

Raw hybridization array data for breast tumor cell lines and tumor tissues

[illegible]

2. InfoVis Introduction

The birth of **Information Visualization = InfoVis**

- Visualization is a **method** of computing. It **transforms** the **symbolic** into the **geometric**, enabling researchers to **observe** their simulations and computations. Visualization offers a method for **seeing the unseen**. It **enriches** the process of scientific discovery and fosters profound and unexpected **insights**. In many fields it is already **revolutionizing** the way scientists do science.

*(**key** journal!) McCormick et al. 1987*

- “...**the biggest problem we face is that of visualization...**”

Richard P. Feynman (Los Alamos, 1945)

Information Visualization

Definition - Use of computer-supported interactive **visual** representation of **abstract** data to amplify **cognition**

- **Compact graphical** presentation
 - Large number of items
 - Extracted from even larger dataset
- **Enable** users to make:
 - Discoveries, Decisions, Explanations
- **Relates** to:
 - Patterns (trends, clusters, gaps, outliers)
 - Groups, Individual items

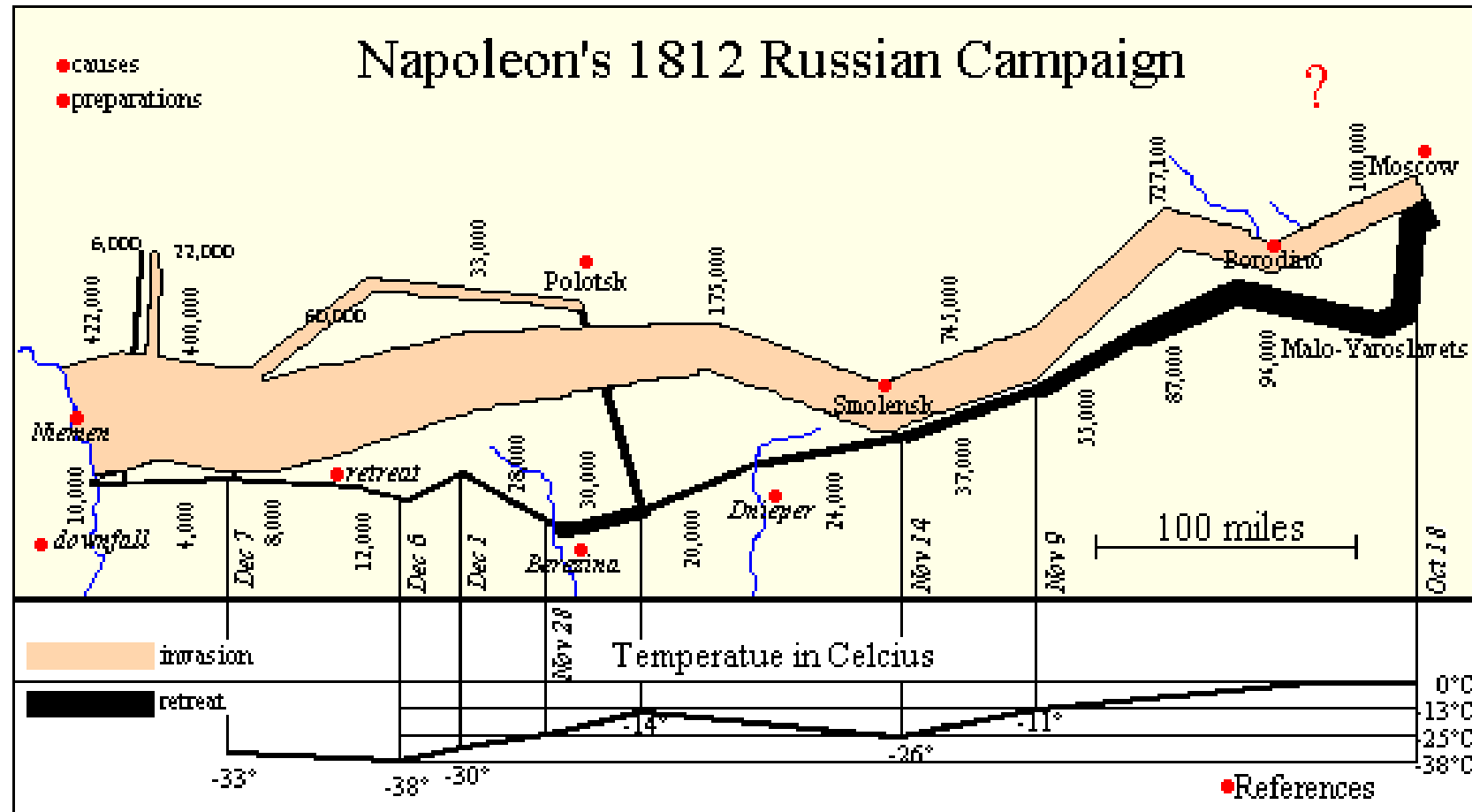
Is it any good?

- Recently, I was asked the following very relevant question, by an eminent Neuroscientist:

“So what has IV ever done for anyone? “

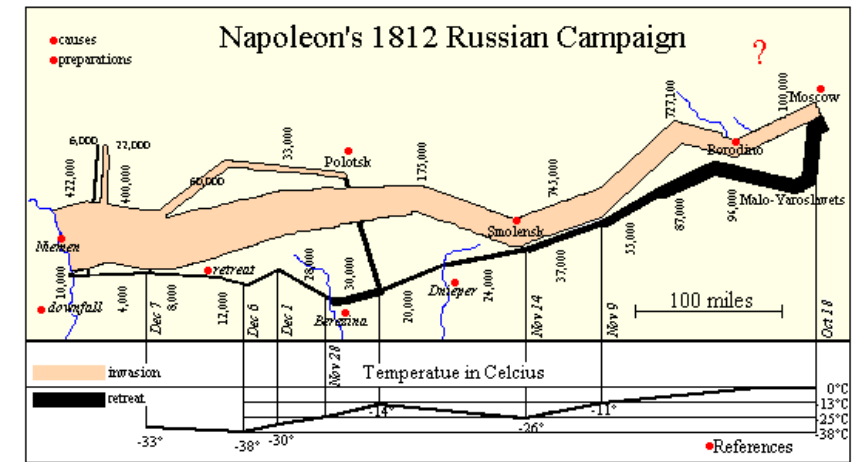
- Of course, the answer is **“quite a lot actually”** but its not always glaring you in the face - unlike significant advances in brain theory for example.
- Well, back in 1932 we had the following ...

Minard's Map – 1812-1813

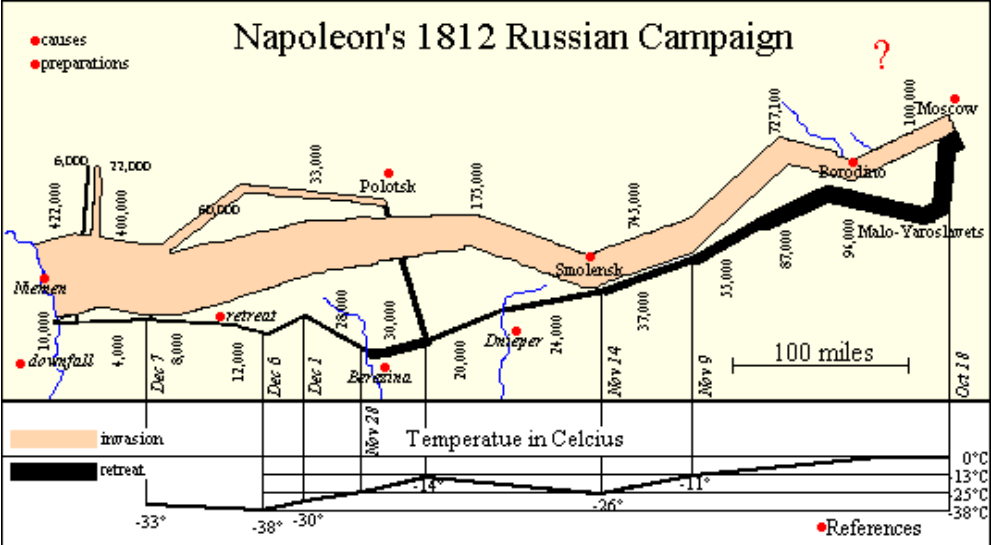


Napoleon's March 1812 - 1813

- Napoleon had a huge army for his Russian campaign, approx. 422,000 men.
- Napoleon did not realize the difficulties of feeding such a big army without proper provisions.
- Napoleon's goal was to live off the land/country => not enough food for them!
- By the time the army reached Moscow, there were only 100,000 men left and only about 10,000 made it back home.
- The rest had either died from hunger or diseases or had been left behind because they were wounded or sick.

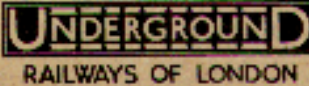


Napoleon's March 1812 - 1813

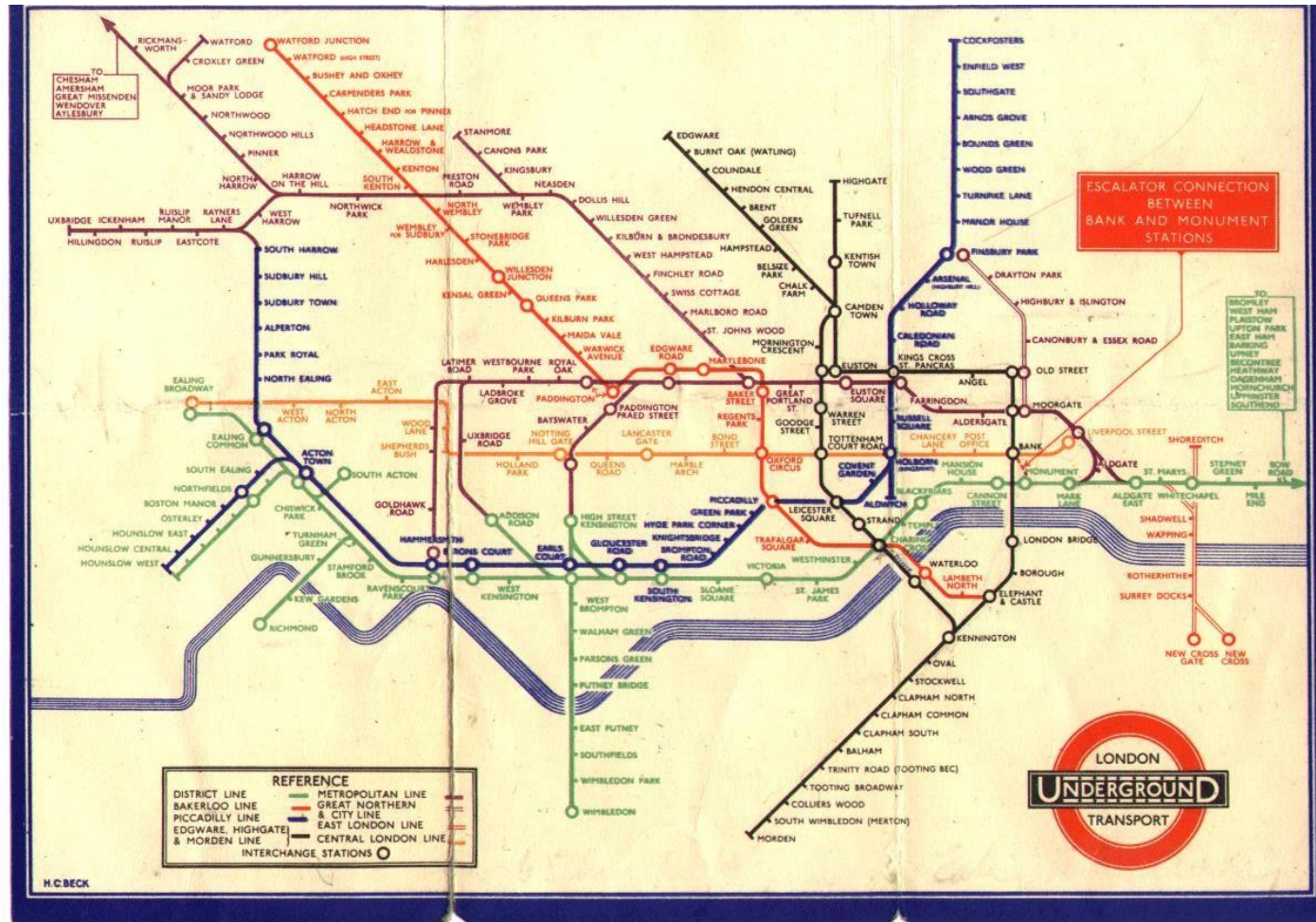


This flow map is famous for the **amount of information** it conveys in such a **limited space**. It plots several variables. the **size of the French army** depicted by the **width** of the bands;

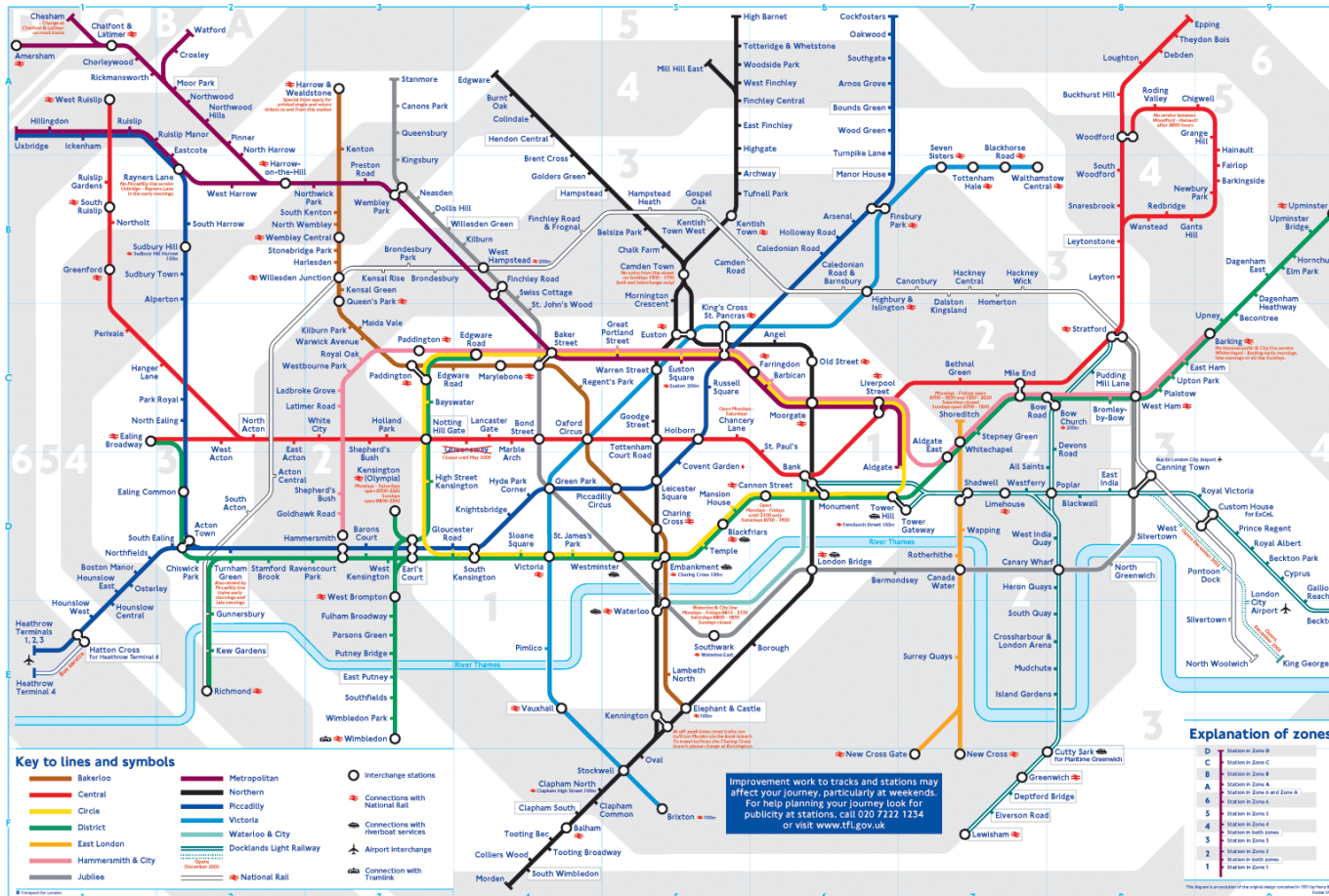
- its **location** on a two-dimensional surface;
- the **direction** of the movement of the advance (upper band) and retreat (lower band);
- the **temperature** on certain dates during the retreat



The Tube Map – 1933

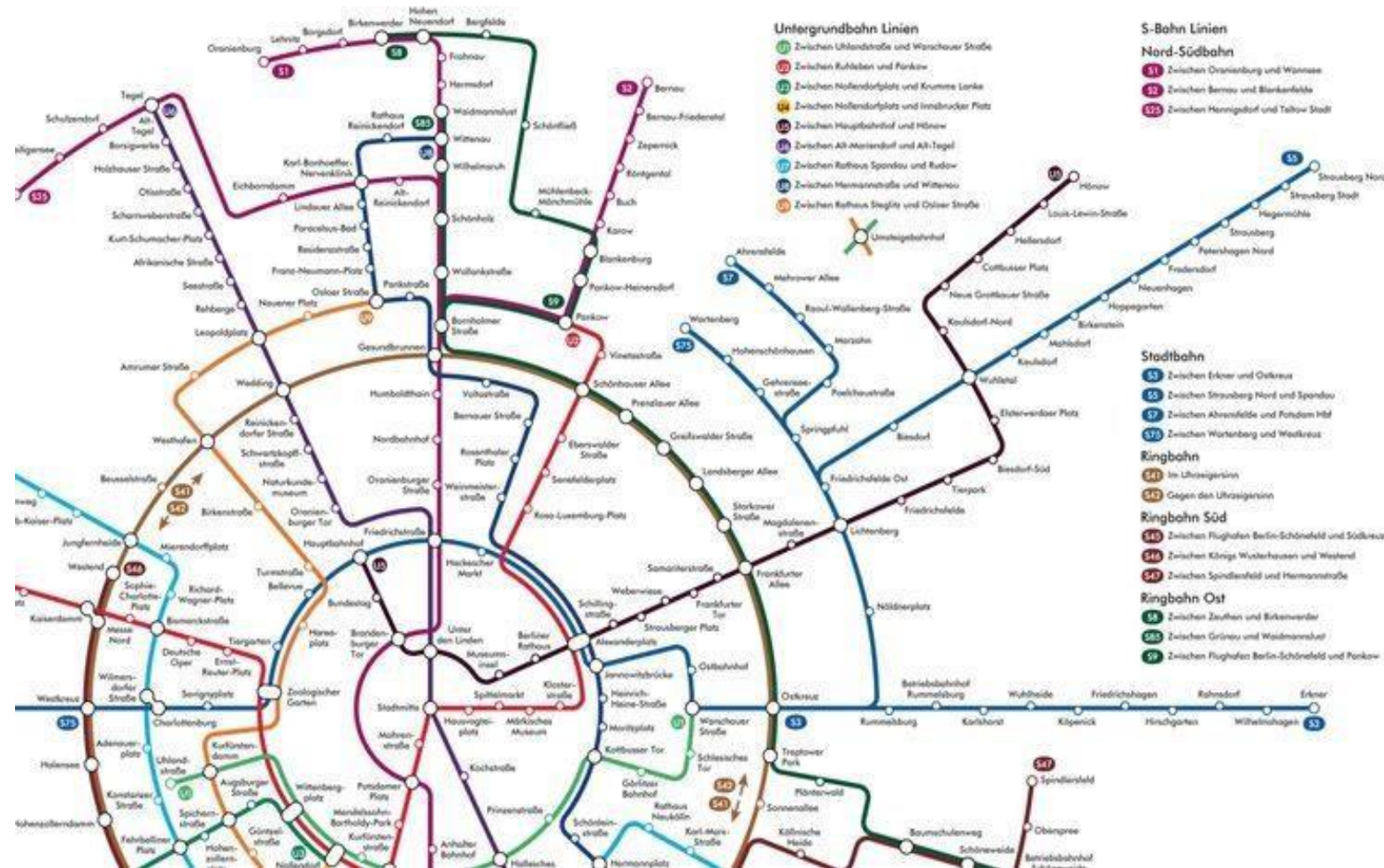


The Tube Map - today



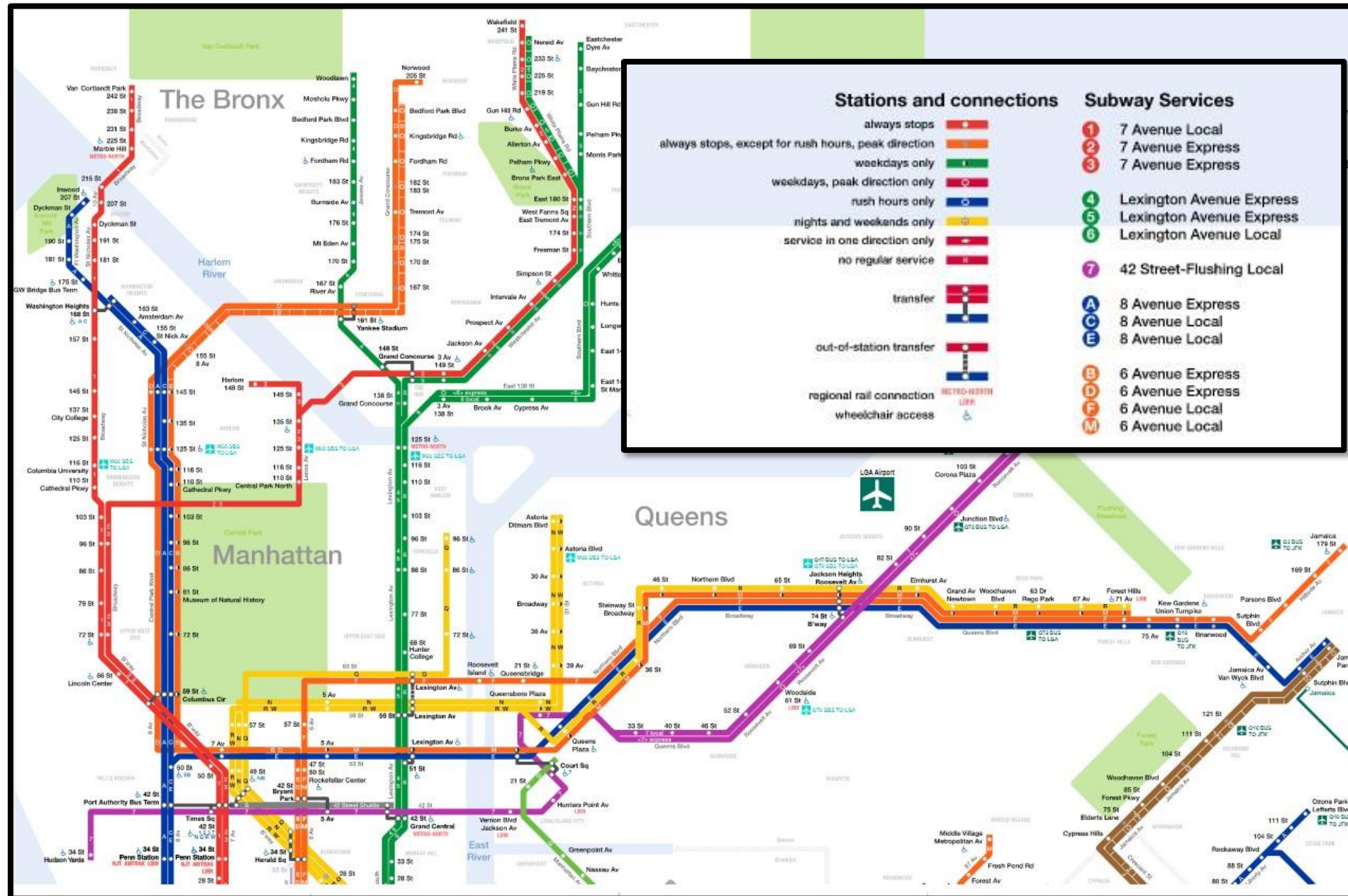
Check out others ... (circular based)

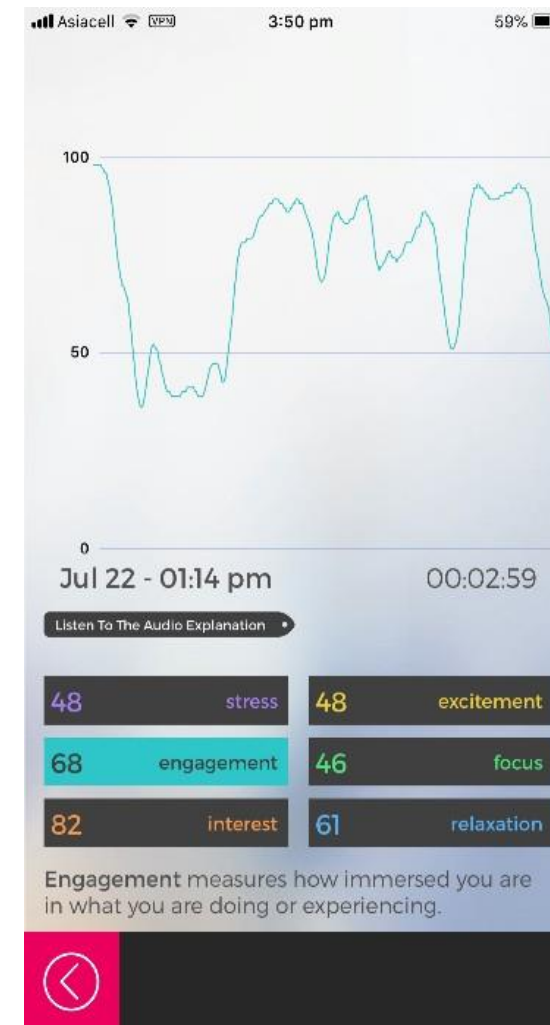
<https://www.intelligenttransport.com/transport-articles/71140/subway-metro-map-user-experience/>



Other options ... <http://www.tubemapcentral.com/showcase/favourites.html>

NYC map - its more like an infographic - lots more data





Data Types

- **Qualitative**

- Nominal have **no natural order** like Male, Female, Others
- Ordinal **have natural order** like A, B C

- **Quantitative**

- Discrete based (Integers) like number of children
- Continuous (Floats, Singles, Doubles) like the length of a piece of string

- **Examples**

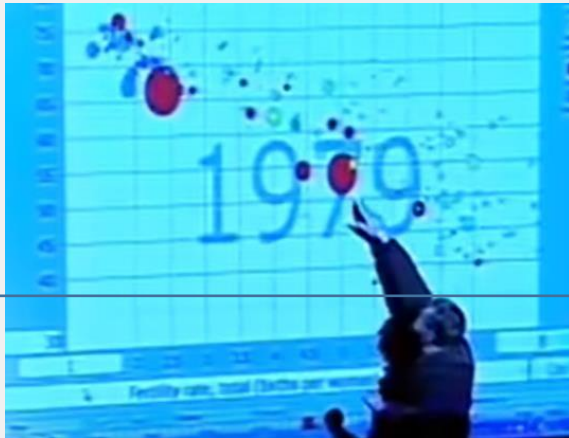
- 1D Linear, 2D Map, 3D World, Multi-Dim, Temporal, Tree, Network
- Require different considerations when representing

<https://www.upgrad.com/blog/types-of-data/#Qualitative-Data-Type>

GapMinder by Hans Rosling



- 00:00 – 20:34
- STOP @ 08:42



https://www.youtube.com/watch?v=hVimVzgtD6w&ab_channel=TED

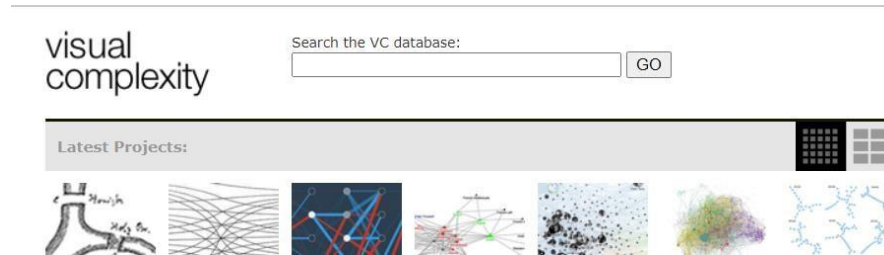
Browsing HCIL

HCIL Overview



- <https://hcil.umd.edu/hcil-overview/>

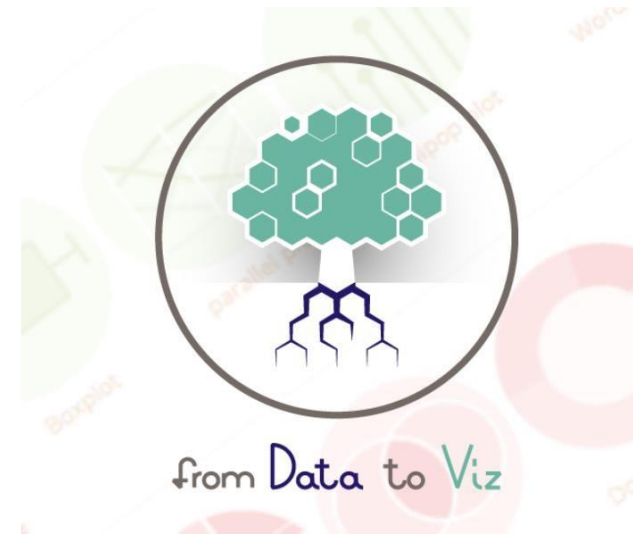
Browsing visual complexity



- <http://www.visualcomplexity.com/vc/>

Browsing data to viz

- <https://www.data-to-viz.com/>



Thank you